Yellow fever vaccine recommendations

Yellow fever vaccine is recommended for people aged 9 months or older who are traveling to or living in areas at risk for yellow fever virus transmission in South America and Africa. Yellow fever vaccine may be required for entry into certain countries. Yellow fever vaccination requirements and recommendations for specific countries are available on the CDC Travelers' Health page.

Serious adverse events can occur following yellow fever vaccination. Therefore, persons should only be vaccinated if they are at risk of exposure to yellow fever virus or require proof of vaccination for country entry.

To minimize the risk of serious adverse events, healthcare providers should carefully observe the contraindications and consider the precautions about vaccination prior to vaccine administration (see below). A medical waiver can be given for persons with a precaution about or contraindication to vaccination. More information about medical waivers is available on the CDC Travelers' Health website.

For more information about the use of yellow fever vaccine in travelers or laboratory workers, see the Advisory Committee on Immunization Practice (ACIP) recommendations.

This vaccine is administered only at designated vaccination centers. For healthcare professionals who want to learn more about yellow fever disease and vaccine, a training module is available.

Yellow fever vaccine virus has been transmitted through blood transfusion and solid organ transplantation. Healthcare providers who administer yellow fever vaccine should inform patients of the need to defer blood donation for at least 2 weeks after receiving the vaccine.

Booster doses of the vaccine

In February 2015, the CDC Advisory Committee on Immunization Practices (ACIP) approved a new recommendation that a single dose of yellow fever vaccine provides long-lasting protection and is adequate for most travelers. The updated recommendations also identify specific groups of travelers who should receive additional doses and others for whom additional doses may be considered including:

- Women who were pregnant when first vaccinated
- Persons who received a hematopoietic stem cell transplant following their last dose of yellow fever vaccine
- Persons who are HIV-infected

- Travelers who received yellow fever vaccine at least 10 years previously and who will be in a higher-risk setting based on season, location, activities, and duration of their travel
- Laboratory workers who routinely handle wild-type yellow fever virus

The official ACIP recommendations were published on June 19, 2015 (see Advisory Committee on Immunization Practice (ACIP) recommendations. All current ACIP yellow fever vaccine recommendations can be found on the ACIP website.

Because of the change to International Health Regulations no longer requiring booster doses, a completed International Certificate of Vaccination or Prophylaxis is now valid for the lifetime of the vaccinee. For more information on completing the International Certificate of Vaccination or Prophylaxis, visit the most recent version of *CDC Yellow Book: Health Information for International Travel*.

Contraindications

- Allergy to a vaccine component
- Aged younger than 6 months
- Symptomatic HIV infection or CD4+ T-lymphocytes <200/mm3 (<15% of total in children aged <6 years)
- Thymus disorder associated with abnormal immune function
- Primary immunodeficiencies
- Malignant neoplasms
- Transplantation
- Immunosuppressive and immunomodulatory therapies

Precautions

- Age 6 to 8 months
- Age 60 years or older
- Asymptomatic HIV infection and CD4+ T-lymphocytes 200 to 499/mm3 (15-24% of total in children younger than 6 years old)
- Pregnancy
- Breastfeeding

Coadministration with other vaccines

Inactivated vaccines

Inactivated vaccines can be administered either simultaneously or at any time before or after yellow fever vaccination.

Live attenuated viral vaccines

Generally, yellow fever vaccine can be given at the same time as other live viral vaccines. If simultaneous administration is not possible, wait 30 days between vaccinations, because the immune response to a live viral vaccine could be impaired if it is administered within 30 days of another live viral vaccine.

There are several studies on the immunogenicity of yellow fever and measle-containing vaccines when coadministered or separated at different intervals. Although coadministration typically results in similar seroconversion rates compared to administered at least 4 weeks apart, the concentration of antibodies in the blood seen following coadministration is lower for all components other than measles. These findings suggest that it is best to give yellow fever and MMR (measles, mumps, and rubella) vaccines 30 days apart. However, coadministration of yellow fever and MMR vaccines is acceptable to ensure travelers are vaccinated before departure.

Live attenuated bacterial vaccines

Limited data suggest oral typhoid vaccine (Ty21a) and oral cholera vaccine (CVD103-HgR) can be administered simultaneously or at any interval before or after yellow fever vaccine.

Reactions to yellow fever vaccine

Reactions to yellow fever vaccine are generally mild and include headaches, myalgias, and low-grade pyrexia. There have been reports of rare but serious events following yellow fever vaccination. These events include anaphylaxis, yellow fever vaccine-associated viscerotropic disease (YEL-AVD), and yellow fever vaccine-associated neurologic disease (YEL-AND). CDC can provide laboratory testing support for suspected YEL-AVD and YEL-AND cases.

Healthcare providers are encouraged to report all adverse events that might be caused by vaccination to the CDC/FDA Vaccine Adverse Events Reporting System (VAERS) by one of the following methods:

1. Submit a report online
2. Print a VAERS form and upload as instructed.